

Hongwei Li

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1 Research Interests

My research centers around **AI agents for Security and SE and agentic Fine-tuning**. My current research focuses on developing customized *AI agents systems* and *reasoning LLMs* for software security and engineering tasks such as patch generation, vulnerability detection, and PoC generation. I also enjoy playing CTF as a core member of Shellphish, a renowned CTF team.

2 Highlighted Projects & Skills

Advanced LLM agents for software engineering and security

- Design and develop pioneer LLM agents with *domain-specific tools, advanced workflows, and memory managements* for SE and security
- Outcomes:
 - **OpenSage** — A self-programming agent generation engine, ranked Top-1 agent on Terminal-Bench, accepted at ICML 2026;
 - **CyberSage** — A fully autonomous security agent built on OpenSage, solved 7 challenges in DEF CON CTF Qualifiers 2026;
 - **PatchPilot** — A cost-efficient issue resolving agent, ranked Top-5 tools on SWE-bench-Verified;
 - **Artiphishell** — A multi-agent system for vulnerability detection and patching, ranked Top-5 in DARPA AIXCC competition;
- Skills: Google Agent Development Kit, Agent design, MCP protocols, Fuzzing (Generic/Directed /Kernel/Web3), Static Analysis(CodeQL/Joern), Concolic Execution(Angr), Reverse Engineering, Vulnerability Exploitation

LLM post-training and agentic training

- Design *novel training recipes for distillation with agentic data*; data selection strategies, running multiple teacher models in our domain-specific agents, unique data filtering and constitution-based data rationalization and correction, and SFT-based fine-tuning; the obtained small models outperform large teacher models in our target tasks
- Outcomes:
 - **VulnLLM-R** - The first reasoning-based large language model for vulnerability detection, equipped with agent scaffolding and agentic fine-tuning, outperforms SOTA tools;
 - **Co-PatcheR** — A set of small reasoning models for issue resolving, ranked Top-2 open-weight models on SWE-bench-Verified;
- Skills: LLaMA-Factory, SFT, vLLM, RL, PyTorch, HuggingFace, verl, Data/model/pipeline parallelism (e.g., FSDP, DDP)

3 Education

- University of California, Santa Barbara**, Goleta, CA Sep. 2024 – Present
Ph.D. Student, Department of Computer Science, advised by Prof. Wenbo Guo
- University of California, Berkeley**, Berkeley, CA Feb. 2026 – Mar. 2026
Visiting Affiliate Researcher, Department of Computer Science, hosted by Prof. Dawn Song
- Purdue University**, West Lafayette, IN Sep. 2023 - May. 2024
Ph.D. Student, Department of Computer Science, advised by Prof. Wenbo Guo
- Shanghai Jiao Tong University**, Shanghai, China Aug. 2020 - Mar. 2023
M.Eng. in Cybersecurity, advised by Prof. Yue Wu
- Shanghai Jiao Tong University**, Shanghai, China Sep. 2016 - Jun. 2020
B.Eng. in Information Engineering; B.A. in French (Dual Degree)

4 Publications

LLM post-training and agents

1. **Hongwei Li**, Ziyue Pan, Zhun Wang, Junmin Zhu, Tianneng Shi, Saastha Vasan, Yuzhou Nie, Christopher Kruegel, Giovanni Vigna, Dawn Song, Wenbo Guo, “CyberSage: Next Generation AI Agents for Cyber Security”, Under Review, 2026
2. **Hongwei Li**, Spandan Garg, Yufan Huang, “Improving Large Language Models for Code through Runtime Program-State Reasoning”, Under Review, 2026
3. Zhun Wang, Nico Schiller, **Hongwei Li**, Srijiith Sessa Narayana, Milad Nasr, Nicholas Carlini, Xiangyu Qi, Eric Wallace, Elie Bursztein, Luca Invernizzi, Kurt Thomas, Yan Shoshitaishvili, Wenbo Guo, Jingxuan He, Thorsten Holz, Dawn Song, “ExploitGym: Can AI Agents Turn Security Vulnerabilities into Real Attacks?”, Under Review, 2026
4. [ICML’26] **Hongwei Li**[†], Zhun Wang[†], Qinrun Dai, Yuzhou Nie, Jinjun Peng, Ruitong Liu, Jingyang Zhang, Kaijie Zhu, Jingxuan He, Lun Wang, Yangruibo Ding, Yueqi Chen, Wenbo Guo, Dawn Song, “OpenSage: Self-programming Agent Generation Engine”, In Proceedings of *The 43rd International Conference on Machine Learning*, Seoul, KR, 2026
[†] These authors contributed fully equally as co-first authors; author order does not reflect contribution
5. Yuzhou Nie, **Hongwei Li**, Ruizhe Jiang, Chengquan Guo, Zhun Wang, Bo Li, Dawn Song, Wenbo Guo, “VulnLLM-R: Specialized Reasoning LLM with Agent Scaffold for Vulnerability Detection”, Under Review, 2026
6. Shellphish, “An AI-powered Cyber Reasoning System for Automatic Vulnerability Identification and Patching”, Under Review, 2026
7. Tianneng Shi, Jingxuan He, Zhun Wang, Linyu Wu, **Hongwei Li**, Wenbo Guo, Dawn Song, “Progent: Programmable Privilege Control for LLM Agents”, Under Review, 2026

8. Yujin Potter, Wenbo Guo, Zhun Wang, Tianneng Shi, **Hongwei Li**, Andy Zhang, Patrick Gage Kelley, Kurt Thomas, Dawn Song, "SoK: Frontier AI's Impact on the Cybersecurity Landscape", Under Review, 2026
9. Zhun Wang, Srijiith Sessa Narayana, Hao Wang, Patrick Großmann, **Hongwei Li**, Nico Schiller, Milad Nasr, Nicholas Carlini, Xiangyu Qi, Eric Wallace, Yan Shoshitaishvili, Wenbo Guo, Jingxuan He, Thorsten Holz, Dawn Song, "How do AI Agents Exploit Security Vulnerabilities?", Under Review, 2026
10. Yuzhou Nie, Ruilin Zhou, Zhaorun Chen, Jingyang Zhang, Yan Shao, **Hongwei Li**, Bo Li, Dawn Song, Wenbo Guo, "NitroBox: Lightning-Fast Sandbox for Large-Scale RL Training", Under Review, 2026
11. Bangya Liu, Jianhong Tu, Yueyang Pan, Hanhong Zhao, Zhe Ye, **Hongwei Li**, Saastha Vasan, Katherine Lambert, Junmin Zhu, Hao Wang, Jienan Chen, Yunze Xiao, Minglai Yang, Zihan Wang, Hairong Yin, Hao Wang, Zhiwen Fan, Tianneng Shi, Zhun Wang, Chenguang Wang, Dawn Song, "CyberArena: Benchmarking Coding Agents in Live Attack-and-Defense CTF", Under Review, 2026
12. Lambang Akbar Wijayadi, Zhaoqi Xiao, Zhun Wang, **Hongwei Li**, Heng Yin, Zhenkai Liang, Dawn Song, "CyberBinGym: Evaluating AI Agents on Binary-Only Vulnerability Analysis and Exploit Generation", Under Review, 2026
13. [ICLR'26] Yuheng Tang, Kaijie Zhu, Bonan Ruan, Chuqi Zhang, Michael Yang, **Hongwei Li**, Suyue Guo, Tianneng Shi, Zekun Li, Christopher Kruegel, Giovanni Vigna, Dawn Song, William Yang Wang, Lun Wang, Yangruibo Ding, Zhenkai Liang, Wenbo Guo, "DevOps-Gym: Benchmarking AI Agents in Software DevOps Cycle", In Proceedings of *The 14th International Conference on Learning Representations*, Rio de Janeiro, Brazil, 2026
14. [ICML'25] **Hongwei Li**, Yuheng Tang, Shiqi Wang, Wenbo Guo, "PatchPilot: A Verifiable and Cost-Efficient Agentic Patching Framework", In Proceedings of *The 42nd International Conference on Machine Learning*, Vancouver, CA, 2025
15. [NeurIPS'25] Yuheng Tang, **Hongwei Li**, Kaijie Zhu, Michael Yang, Yangruibo Ding, Wenbo Guo, "Co-Patcher: Collaborative Software Patching with Component-specific Small Reasoning Models", In Proceedings of *The 39th Conference on Neural Information Processing Systems*, San Diego, CA, 2025

RL and DNN for Security

1. Wenxuan Shi, **Hongwei Li**, Jiahao Yu, Xinqian Sun, Wenbo Guo, Xinyu Xing, "BandFuzz: An ML-powered Collaborative Fuzzing Framework.", arXiv preprint arXiv:2507.10845.
2. [ICSE SBFT'24] Wenxuan Shi, **Hongwei Li**, Jiahao Yu, Wenbo Guo, Xinyu Xing, "BandFuzz: A Practical Framework for Collaborative Fuzzing with Reinforcement Learning", In Proceedings of *International Workshop on Search-Based and Fuzz Testing*, Lisbon, Portugal, 2024
3. [Computers & Security'22] Jingcheng Yang, **Hongwei Li**, Shuo Shao, Futai Zou, Yue Wu, "FS-IDS: A framework for intrusion detection based on few-shot learning", *Computers & Security*, 122: 102899, 2022.

5 Selected Competitions

- Terminal-Bench, 2026 (**Top 1 Agent**) (OpenSage)
- AIXCC, DARPA, 2023-2025 (**5th Place; Awarded \$3M**)
- SWE-Bench patching leaderboard, 2025 (**Top 5 tools; Top 2 Open-weight models**)
- Google SBFT Fuzzing Tool Competition, 2024 (**Top 1**)
- DEF CON CTF Qualifiers, 2026 (**45th Place**) (CyberSage, fully automated, 7 challenges solved)
- DEF CON CTF Final, 2025 (**11th Place**), 2026 (**10th Place**)
- SunshineCTF, Sep 2021 (**11th Place**) (*Solo Team in a Team-based CTF*)